

Synoptic CB: Macrophyte Fluxes

2022 Samples

2026-03-04

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#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("ch4_", .)
  ) %>%
  rename(ch4_qaqc_flag = qaqc_flag)

co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("co2_", .)
  ) %>%
  rename(co2_qaqc_flag = qaqc_flag)

# Combine all files
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_
```

##Remove bad fluxes and calculate how many there were

Number flux measurements removed: 2

Number negative CO2 fluxes removed: 0

Number messy fluxes removed: 2

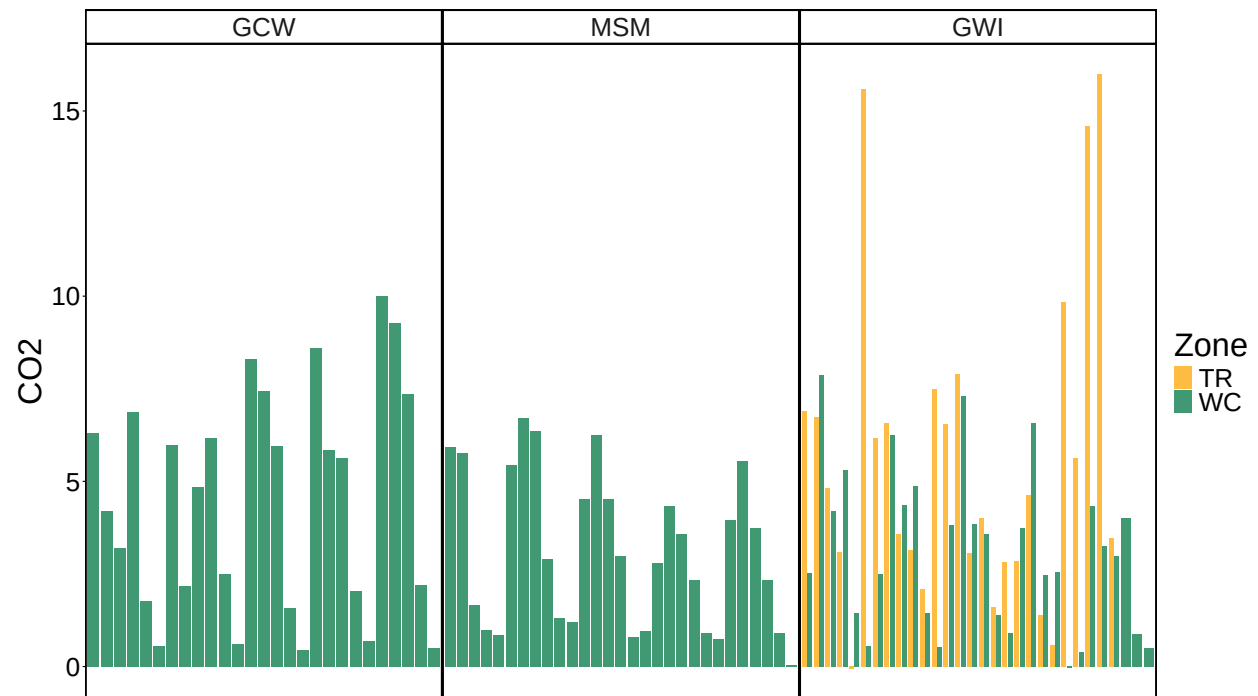
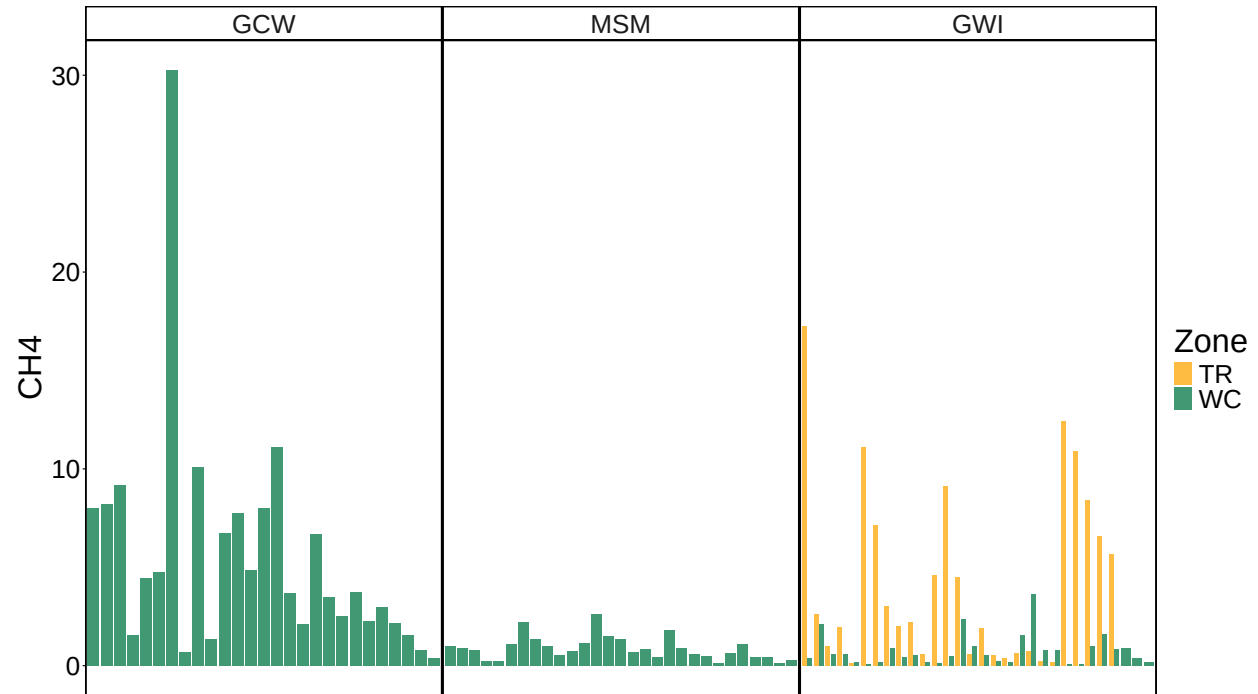
Percentage flux measurements removed: 1.74%

Percentage negative CO2 fluxes removed: 0%

Percentage messy fluxes removed: 1.74%

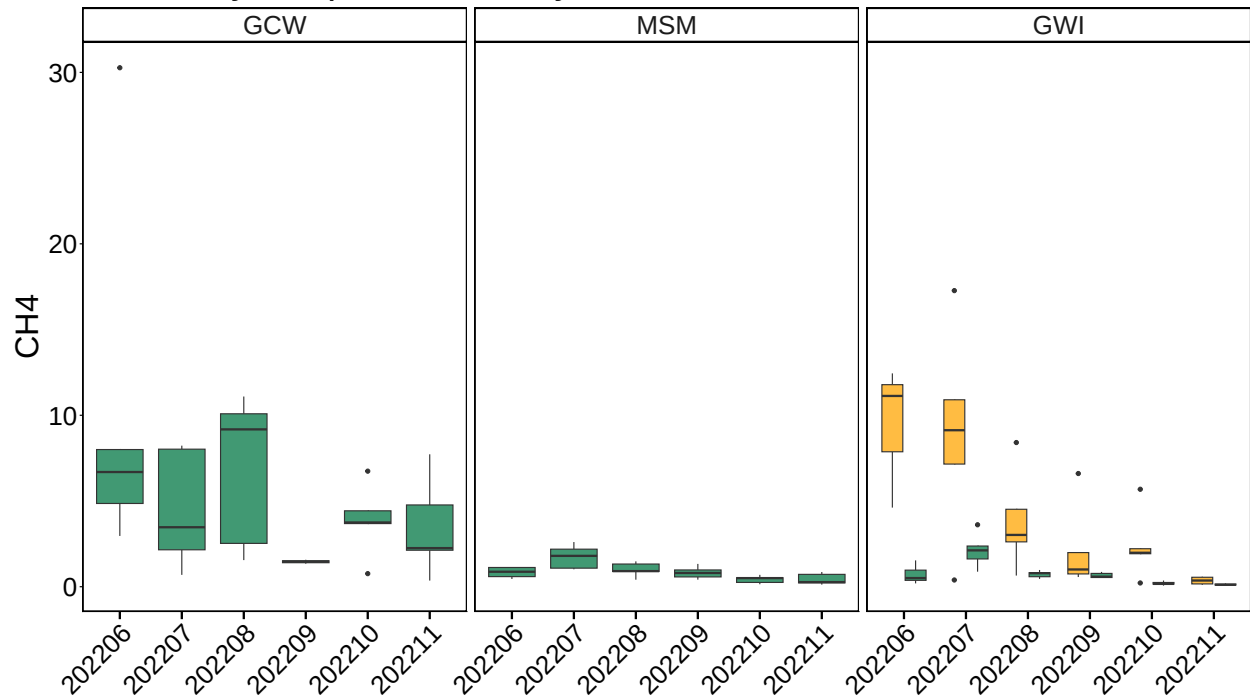
0.1 Write out files

0.2 All Data by Year, Site and Zone

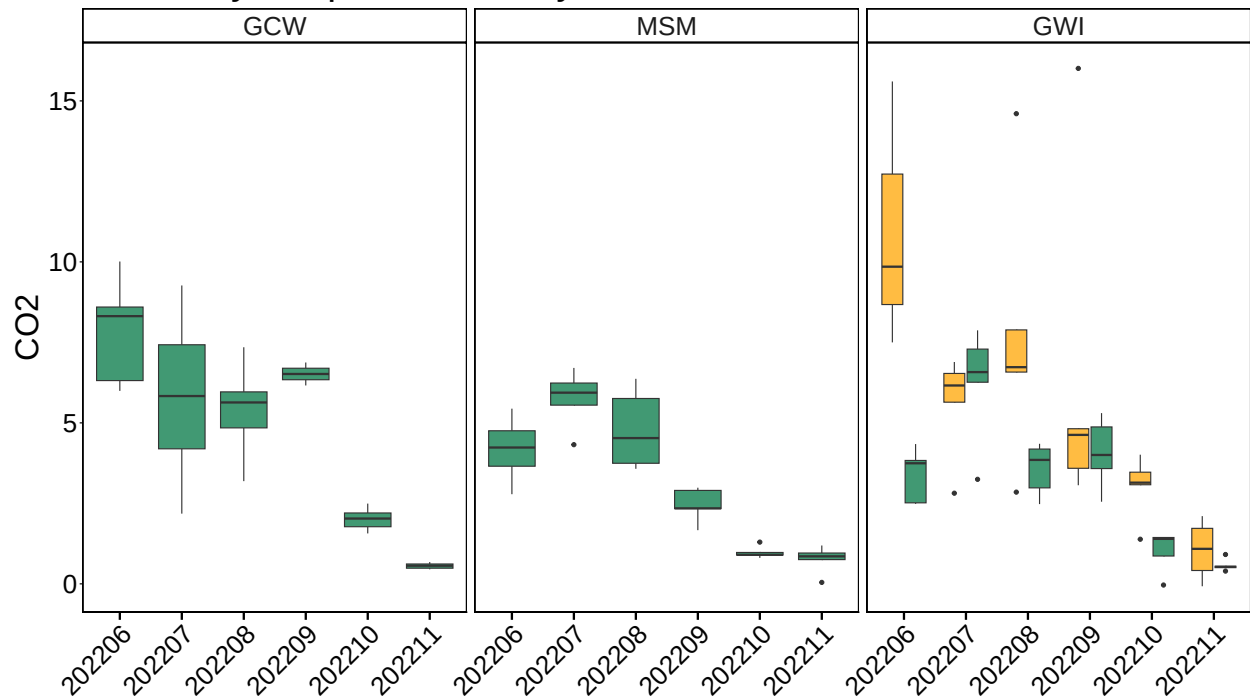


0.3 Boxplot summary by Date

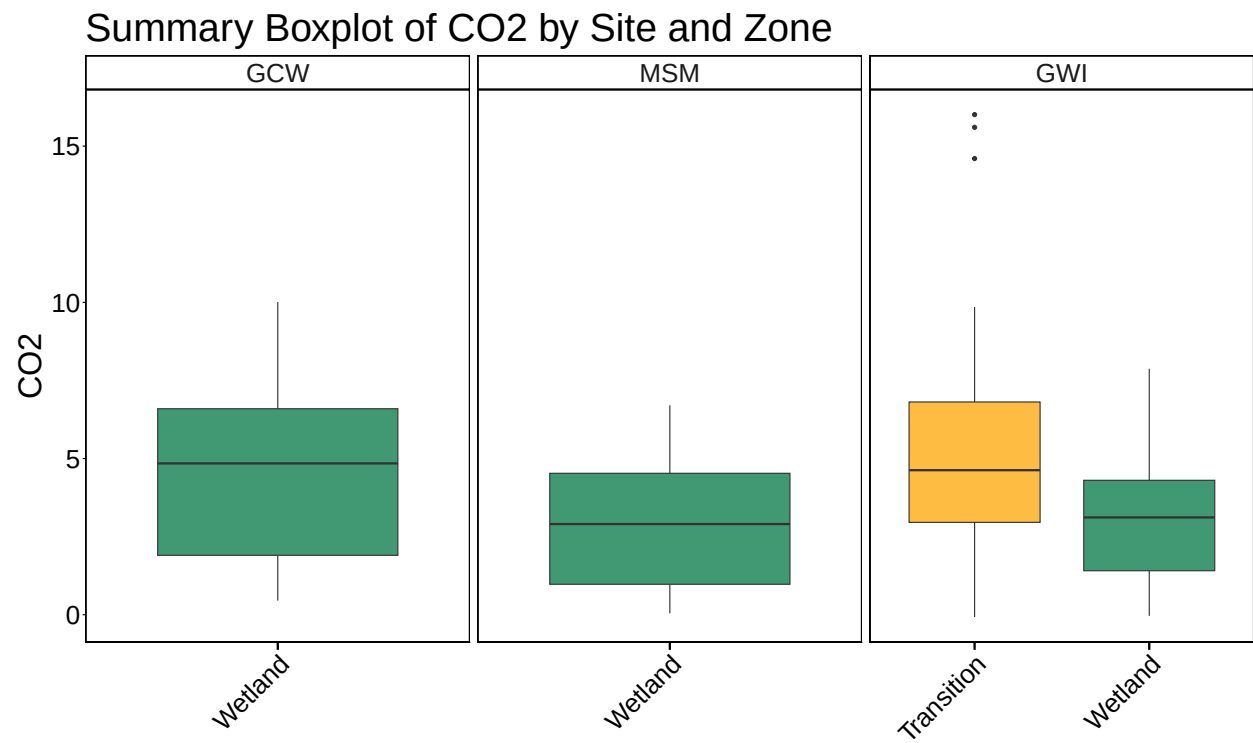
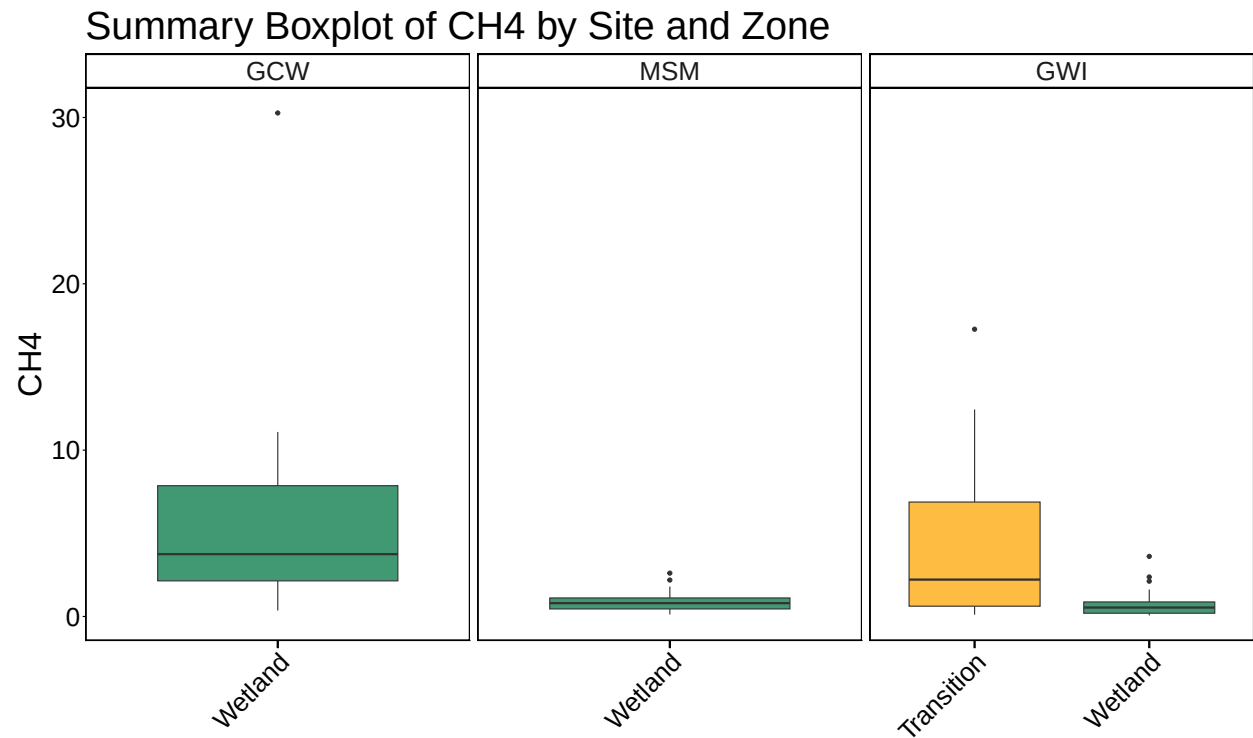
Summary Boxplot of CH4 by Site and Zone



Summary Boxplot of CO2 by Site and Zone



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
GCW	Wetland	0.3585206	30.270376	5.5296620	0.4518196	10.011470	4.481874
MSM	Wetland	0.1222875	2.606256	0.8783515	0.0458841	6.706775	3.115307
GWI	Transition	0.1071858	17.271153	4.3189199	-0.0745288	16.009112	5.594259
GWI	Wetland	0.0653984	3.612942	0.7590672	-0.0364413	7.873652	3.144701